

PITP Mobile App Development Outline

This course is structured to take **absolute beginners** step-by-step from basic programming concepts to building a complete Mobile App using Flutter.

Phase 1: Programming Fundamentals with Dart

Introduction to Programming

- What programming is and how it works
- Understanding how computers execute instructions
- Real-world examples of programming

Dart Language Basics

- What is Dart, and where is it used?
- Why Dart is the language of Flutter
- Using DartPad (online editor)
- Installing Dart SDK (optional)

Dart Program Structure

- `main()` function
- Code blocks and syntax rules
- Writing and running a Dart program

Output & Comments

- Printing text and numbers
 - Single-line and multi-line comments
 - Understanding program execution flow
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Phase 2: Variables, Data Types & Operators

Variables & Data Types

- What are variables, and why are they needed
- `int`, `double`, `String`, `bool`
- Naming rules and best practices

Type Inference

- Using `var`
- Using `dynamic`
- Difference between explicit and inferred types

Operators

- Arithmetic operators (+, -, *, /, %)
- Relational operators (>, <, ==, !=)
- Logical operators (&&, ||, !)
- Assignment operators (+=, -=, *=)

Practice & Logic Building

- Simple calculation programs
 - Comparison-based problems
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Phase 3: Input, Control Flow & Logic

Input & Output

- Console output
- Introduction to taking input (conceptual understanding)

String Interpolation

- Using \$ and \${ }
- Combining text with variables

Math Utilities

- Basic math operations
- Using common math functions

Control Flow

- if statements
 - if-else conditions
 - else-if ladders
 - Writing decision-based programs
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Phase 4: Loops & Functions

Loops

- for loop
- while loop
- do-while loop
- Loop use-cases and common mistakes
- Pattern-based and counting programs

Functions

- What are functions and why are they used
 - Creating and calling functions
 - Parameters and arguments
 - Return types
 - Arrow (short-hand) functions
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Phase 5: Collections (Data Structures)

Lists

- Creating lists
- Accessing list elements
- Common list methods
- Iterating over lists

Sets

- What is a set
- Difference between list and set
- Removing duplicates

Maps

- Key-value structure
 - Creating and accessing maps
 - Common map operations
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Phase 6: Object-Oriented Programming (OOP) in Dart

Classes & Objects

- What is OOP
- Creating classes
- Creating objects
- Constructors

Inheritance

- Parent and child classes
- Reusing code using inheritance
- Method overriding

Encapsulation

- Private variables
- Getters and setters
- Data protection

Static Members

- Static variables
- Static methods

Exception Handling

- Understanding errors
 - Try-catch blocks
 - Handling runtime exceptions
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Phase 7: Flutter Fundamentals

Introduction to Flutter

- What is Flutter
- Flutter architecture
- Difference between Flutter and native apps

Environment Setup

- Installing Flutter SDK
- Android Studio & VS Code setup
- Creating the first Flutter project

Project Structure

- Understanding folders and files
 - Entry point of Flutter app
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Phase 8: Widgets & UI Basics

Widgets Concept

- What are widgets
- Stateless widgets
- Stateful widgets
- Hot reload vs hot restart

Material Design Basics

- MaterialApp & Scaffold
- Text, Container
- Icon and Image widgets

Layout Widgets

- Row
- Column
- Stack
- MainAxis & CrossAxis alignment

Buttons & Interaction

- ElevatedButton
- TextButton
- IconButton

First Project: WhatsApp Clone

Phase 9: Navigation & UI Components

Navigation

- Navigator push & pop
- Named routes

Lists & Grids

- ListView
- ListView.builder
- GridView

Custom Widgets

- Breaking UI into reusable widgets

Forms & User Input

- TextField
- Controllers
- Form validation

Dialogs & Bottom Sheets

- AlertDialog
 - BottomSheet
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Phase 10: API Integration & State Basics

REST API Fundamentals

- What is an API
- HTTP methods (GET, POST)

API Integration

- Using the HTTP package
- Fetching data from the API
- Handling loading, success, and error states

JSON Handling

- Understanding JSON format
- Parsing JSON
- Creating model classes

State Management (Basic)

- Using `setState()`
- Passing data between widgets

Second Project: News App using Fake API

Phase 11: Local Storage

SharedPreferences

- Saving simple data locally
- Loading stored data

SQLite (Basics)

- What is SQLite
- CRUD operations (Create, Read, Update, Delete)
- Using local database in Flutter

Final Project: Task Manager App

Phase 12: App Publishing & Play Store Setup

App Release Basics

- What is app publishing
- Difference between debug and release builds
- Understanding app versions and build numbers

Android App Signing

- Keystore concept
- Generating a release keystore
- App signing process overview

Play Store Console Overview

- Creating a Google Play Developer account
- Understanding Play Console dashboard
- App creation steps

Store Listing Setup

- App name and package name
- App description (short & full)
- App icon and feature graphic
- Screenshots and promotional assets

Publishing Process

- Uploading APK/AAB file
- Internal testing vs production release
- Reviewing and publishing the app